

Part 1 of 2

How Secure are Your E-mails

Despite all the advice, we treat our e-mails as if they are the same as personal conversation. Private. Off the record. We also assume that no one else is going to be able to read it, and that it cannot ever get into the wrong hands. Slowly but surely we are finding out, the hard way, that, as in the words of the song, 'It ain't necessarily so.' It appears that everyone, from our e-mail administrator to half the hacking community, can pick up what we are doing, even off the internal network



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E-mail security has been a growing concern over the past few years. The average individual, who uses e-mail, naively believes that his e-mail is private and secure but he is wrong, as the electronic world is filled with snoopers who can access all types of data over the network. As the world goes digital, with more and more raw information about individuals available electronically, the need for security increases. Ubiquity and speed of e-mail have made it increasingly effective. So providing reliance over this medium has become an inevitable requirement.

The e-mails of today can be made secure by encrypting the content on an external medium and sending this

as an attachment over the mail. But this increases the size of the mail and makes it inconvenient for the user. There are other systems that provide specific security and are strongly tied to the mail servers and browsers. Enter the answer—secure e-mail (Se-mail). Run it just like an ordinary mail but click on the secure button and you are done. Somehow it seems obvious to anyone that the immense complexity of the computer can be made safe and secure by a single act (the laying on of hands perhaps).

Let us take an analogy that explains a typical e-mail scenario. Suppose a person is in a crowded room and spots a friend to whom he wants to convey an important message, but he does not want to yell. He writes a message on a paper-slip and asks the person next to

him to pass it on, trusting no one will read it on its journey through dozens of hands—hands belonging to people he has never met. What are the odds that the person's friend will be the only one to read the message intended for him or her alone? Untold number of people, including you and me, employ this useful aspect of network and Internet communication to exchange data and documents.

Whilst there are strong legal penalties that deter snooping in the traditional media, such as posted or courier mail, the sheer novelty of computer-aided communication has created gaps in the legal net, allowing for tampering and snooping on others' e-mails or personal information. As the world goes digital, with more and more raw information about individuals available electronically, the need for security increases.

The major problems that ordinary e-mails suffer are data origin authentication and connectionless integrity apart from privacy. Which means that it is quite difficult with the ordinary e-mail to prove that the sender actually wrote the message, or that it has not been tampered with? Here, we will be discussing about the various challenges in e-mail security and try to find out the solution for the same. At the same time, we will discuss a few tips and tricks that can ensure e-mail security.

How e-mail works

Sending an e-mail message is like sending an ordinary letter (not a courier, registered letter or speed post).